

Improving Localization with Localization-guided Structure-from-Motion

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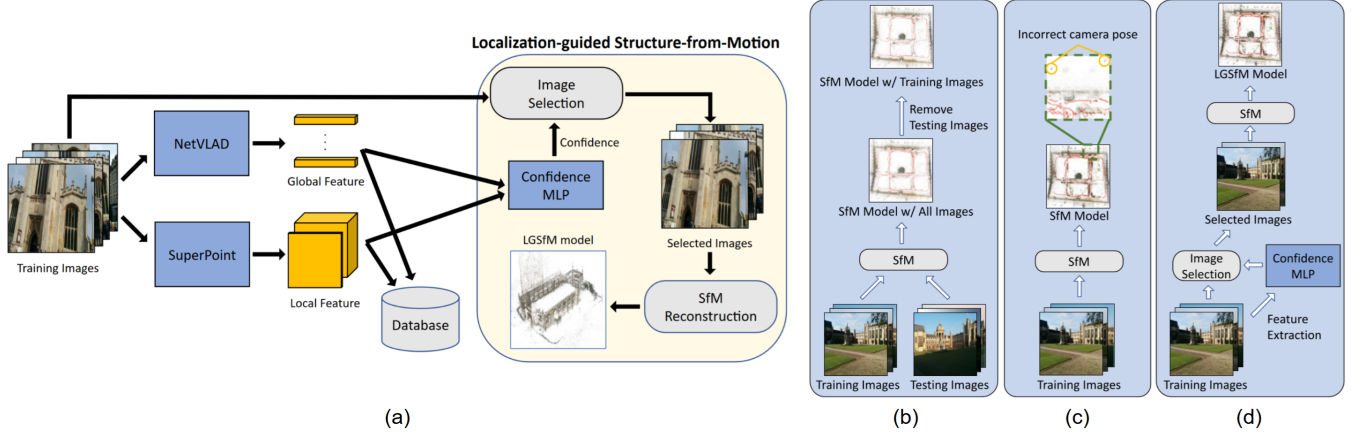


Figure 1: (a) The detailed pipeline of 3D model construction using our method, LGSfM. (b–d) Different strategies for 3D model generation: (b) existing dataset method, (c) real-world localization method, and (d) our proposed method, LGSfM.

Abstract

Most visual localization methods rely on 3D point clouds built with Structure-from-Motion (SfM). However, existing localization datasets often employ SfM to derive camera poses for all images, including testing ones, which does not reflect real-world conditions. Furthermore, SfM is rarely integrated with visual localization, which can limit accuracy. To address this, we introduce Localization-guided Structure-from-Motion (LGSfM), which leverages global and local features to generate pseudo-labels for training a confidence MLP that selects high-confidence images for model construction. This yields a 3D model better optimized for localization. Experiments on the 7-Scenes and Cambridge Landmarks datasets demonstrate that LGSfM outperforms standard SfM, achieving a 44% improvement on the Old Hospital scene at (0.25 m, 2.0°).

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Table 1: Localization success rates of training-only and all-image models. "Hosp." denotes the "Old Hospital" scene.

Scenes	Localization Success (%) ↑ @(0.25m, 2.0°/0.5m, 5.0°)	
	All Images	Training Images
Kings	74.34/90.38	26.53/82.22
Hosp.	65.38/86.81	0/0.55
Shop	95.15/99.03	93.20/99.03
Church	97.36/99.62	72.83/97.74
Court	66.32/82.50	0.13/26.97

1 Introduction

Visual localization estimates a camera's 6-DoF pose from images and is vital for robotics and autonomous driving. Structure-based and hybrid methods rely on 3D point clouds, typically built via Structure-from-Motion (SfM). However, to our knowledge, SfM has not been specifically studied to improve localization performance.

Existing datasets such as Cambridge Landmarks [Kendall et al. 2015] generate camera poses by optimizing over all images, including testing ones, before excluding the testing images from the final model, as illustrated in Fig. 1 (b). This allows testing images to influence the reconstructed 3D model, unlike real-world cases where only training images are available, as shown in Fig. 1 (c). Using hloc [Sarlin et al. 2019], we compare localization performance between

models built from all images and those built from training images only. As shown in Table 1, the latter performs worse, highlighting a discrepancy between experimental and real-world settings.

In this study, we present the first approach to optimize SfM-based 3D models for localization. As shown in Fig. 1 (d), our method uses image features to generate pseudo-labels for training an MLP that predicts image confidence scores. Only high-confidence images are used to build the 3D model. Experiments on 7-Scenes [Shotton et al. 2013] and Cambridge Landmarks [Kendall et al. 2015] show up to a 44% improvement in the number of correctly localized images.

2 Method

2.1 Overview

Fig. 1 (a) shows our framework pipeline. Global and local features are extracted from training images using pretrained NetVLAD [Arandjelovic et al. 2016] and SuperPoint [DeTone et al. 2018]. These features are stored in a database and used by LGSfM to construct the 3D model. Localization is then performed through a hierarchical process [Sarlin et al. 2019].

2.2 Localization-guided Structure-from-Motion

2.2.1 Pseudo-label Generation. To train the confidence MLP, we generate pseudo-labels from global and local features. Images are clustered using Hierarchical DBSCAN [Campello et al. 2013] on global features. For each image I , we retrieve the top- k most similar database images based on global feature similarity and count those from clusters different from I 's, denoted as n_{diff} . If n_{diff} exceeds a threshold τ_{diff} , the pseudo-label of I is set to 0. This removes images with similar global features but irrelevant content, producing a retrieval set better aligned with the query for localization.

We hypothesize that reference images from favorable viewpoints, defined as those with more uniformly distributed keypoints, can improve localization. To quantify this, we compute a keypoint distribution score S for each image by dividing it into 2^m cells and counting those containing keypoints with confidence above τ_{kpt} . Scores are computed across L pyramid layers, weighted by their segmentation level m , and summed to obtain the final S . Higher scores indicate more uniform keypoint distributions. Since keypoint counts depend on image content, scores are compared within clusters, and images are ranked in descending order within each cluster to produce pseudo-labels y :

$$y = \begin{cases} 0, & \text{if } n_{diff} > \tau_{diff} \\ 1 - \frac{rank}{num}, & \text{otherwise,} \end{cases} \quad (1)$$

where $rank$ represents the rank of image I within its cluster and num represents the total number of images in that cluster.

2.2.2 Confidence MLP. We train a 12-layer MLP to predict a confidence score for each image. The input is the global feature concatenated with the flattened positions and confidence of the top 4,096 keypoints. A sigmoid activation outputs a score between 0 and 1, and the network is trained with an L2 loss against pseudo-label y .

2.2.3 Image Selection and SfM. After training, the confidence MLP outputs a score for each image. Images with scores above a threshold τ are used for SfM with COLMAP [Schonberger and Frahm 2016].

Table 2: Localization success rates and median errors on the Cambridge Landmarks [Kendall et al. 2015] dataset, compared with COLMAP [Schonberger and Frahm 2016]. "Hosp." denotes the "Old Hospital" scene.

Scene	Localization Success (%) $\uparrow$ @ (0.25m, 2.0°/0.5m, 5.0°)		Median Errors (m/°) $\downarrow$	
	COLMAP	Ours	COLMAP	Ours
Kings	26.53/82.22	30.03/83.09	0.362/0.333	0.341/0.296
Hosp.	0/0.55	44.51/85.16	1.101/5.138	0.273/1.297
Shop	93.20/ 99.03	94.17/99.03	0.082/0.516	0.066/0.455
Church	72.83/97.74	90.19/98.11	0.198/0.429	0.131/0.352
Court	0.13/26.97	17.50/61.45	0.572/0.242	0.419/0.134

Table 3: Localization success rates and median errors on the 7-Scenes [Shotton et al. 2013] dataset, compared with COLMAP [Schonberger and Frahm 2016]. "Pump." denotes the "Pumpkin" scene, and "Kitch." denotes the "Redkitchen" scene.

Scene	Localization Success (%) $\uparrow$ @ (0.25m, 2.0°/0.5m, 5.0°)		Median Errors (m/°) $\downarrow$	
	COLMAP	Ours	COLMAP	Ours
Chess	25.85/ 99.95	26.40/99.95	0.031/2.378	0.031/2.351
Fire	35.90/99.65	65.10/100	0.028/2.217	0.025/1.741
Heads	36.40/99.40	51.40/100	0.015/2.225	0.015/1.950
Office	26.22/ 96.10	27.22/94.85	0.083/2.913	0.083/2.911
Pump.	2.50/ 93.35	2.80/93.35	0.077/3.312	0.076/3.209
Kitch.	15.30/ 87.88	18.26/87.62	0.062/3.114	0.062/3.029
Stairs	5.10/86.90	7.20/87.30	0.100/3.211	0.076/3.207

3 Experiments and Discussions

We aim to improve visual localization by enhancing 3D reconstruction. We evaluate different reconstruction methods within the same localization pipeline [Sarlin et al. 2019] on the outdoor Cambridge Landmarks [Kendall et al. 2015] and indoor 7-Scenes [Shotton et al. 2013] datasets. Localization relies solely on 3D models built from training images, simulating real-world conditions.

Tables 2 and 3 show that COLMAP [Schonberger and Frahm 2016] performs poorly in several scenes, especially Old Hospital (Hosp.), where median errors exceed 1 m and 5°, revealing the limitations of existing localization methods in real-world settings. By contrast, our method achieves consistent gains across both datasets. In large-scale outdoor environments, Old Hospital (Hosp.) shows the largest improvement, with correct localizations at the strictest threshold increasing by 44%. In indoor scenarios, the Fire scene achieves nearly a 30% improvement under the same conditions.

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